

Community Project
Branch: CSE-CYBERSECURITY & INTERNET OF THINGS(ICB)
Department: Data Engineering



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- **Project Title: VIDYA Uday – DIGITAL EMPOWERMENT AND AI AWARENESS**
- **Batch Number:3**

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Abstract

A concise overview of VidyaUday and its core purpose



Background

- Rural 10th and Intermediate students in Andhra Pradesh & Telangana face an acute guidance deficit, unaware of higher study paths, scholarships, and free learning tools.
- Essential educational information remains fragmented across complex government circulars, English-only websites, and aggressive commercial coaching apps.

Our Solution

VidyaUday provides a 100% free, mobile-friendly platform with zero login, no ads, and full English-Telugu bilingual parity.

- AI Tools Hub: Curated free AI study engines with 1-click copyable student prompts for maths, essays, and vernacular explanations.
- Exclusive Switchers: Interactive career roadmaps (MPC/BiPC/CEC/Poly/ITI), 1-click SCERT board textbooks, and 30-sec scholarship eligibility checks.

Expected Outcome

- Unlocks statutory government scholarships, eliminates cyber café application fees, and fosters confident, self-directed rural students.

Introduction

The challenge rural students face and what VidyaUday does about it



- **The Growing Rural Challenge:** Lakhs of students pass 10th (SSC) and Intermediate (+2) annually without structured counseling, choosing careers blindly based on peer pressure.
- **The Information & Language Barrier:** Genuine resources exist but are buried across multiple portals in dense English. Rural first-generation digital learners struggle to find authentic guidance.
- **Commercial Coaching Exploitation:** Expensive private coaching apps charge ₹20,000–₹50,000, drowning out 100% free Central and State Government learning platforms like DIKSHA and Swayam Prabha.
- **VidyaUday's Integrated Platform:** A lightweight single-page application uniting AI prompts, career roadmaps, board books, verified scholarships, and digital literacy under one roof.
- **Universal Zero-Barrier Accessibility:** Loads in under 1.2s on basic smartphones over 3G/4G, requires no user account, collects zero personal data, and is 100% free forever.



Lakhs of rural 10th and Intermediate students in Andhra Pradesh and Telangana face a severe guidance deficit, making critical career choices blindly without awareness of free government learning resources, AI educational tools, or annual scholarships worth crores of rupees. Scattered, English-dominant government circulars and aggressive commercial coaching traps prevent first-generation learners from accessing opportunities they legally qualify for.

65%**Lack Structured Guidance**

Make higher education choices without knowledge of alternatives or career prospects.

80%**Never Used Free AI**

Unaware of free AI tools like ChatGPT or Gemini for academic homework & study doubts.

₹50K+**Unclaimed Scholarships**

Annual government welfare and DBT scholarship funds missed per eligible rural student.

Project Objectives

Project Objectives

Six clear goals that define what VidyaUday sets out to achieve



- 01. Centralize Educational Access:** Consolidate AI study tools, career roadmaps, SCERT textbooks, and verified scholarships into a single, mobile-optimized web hub.
- 02. Democratize AI for Education:** Equip rural students with 1-click copyable prompts for step-by-step math problem solving, essay writing, and vernacular concept explanations.
- 03. Provide Multi-Stream Career Clarity:** Deliver interactive step-by-step career roadmaps and comparison matrices for MPC, BiPC, CEC, MEC, Polytechnic, and ITI streams.
- 04. Unlock Statutory Government Scholarships:** Provide a 30-second eligibility checker, verified DBT government gazette citations, and direct links to official .gov.in portals.
- 05. Promote Digital & Cyber Safety Literacy:** Educate students on DigiLocker certificate storage, photo/signature compression for exam forms, and fraud awareness against fake scholarship fees.
- 06. Ensure Complete Linguistic Inclusivity:** Provide seamless real-time bilingual switching between English and authentic Telugu (తెలుగు) with zero page reload latency.

Project Scope

Project Scope

What VidyaUday includes — and what it deliberately does not cover



✓ IN SCOPE

- Bilingual web platform for 10th and Inter students in AP & TS.
- Curated free AI tools with 1-click copyable student prompts.
- Interactive roadmaps, comparison matrix, and 30-sec matcher.
- 20+ verified Central & State DBT government scholarships.
- 1-Click SCERT board textbooks & 5-year solved papers hub.
- Digital readiness quiz, exam form guides, and scam shield.

✗ OUT OF SCOPE

- Not a paid college admission or examination booking agency.
- Does not submit official scholarship applications on students' behalf.
- Not a replacement for classroom teachers or school faculties.
- Zero commercial advertisements, paid tiers, or subscription models.
- Does not collect, store, or track student personal data or OTPs.
- Not limited to any single caste, religion, or gender demographic.

Literature Review / Background Study

Existing educational platforms and why they fall short for rural students



Existing Platform	Key Limitation for Rural Students	VidyaUday Advantage
Buddy4Study	Commercial paid tests; no Telugu support; no AI awareness.	100% free guidance, vernacular support & 1-click prompts.
Careers360	Urban-focused; complex, ad-heavy UI; English-only.	Clutter-free single-view UI; authentic Telugu parity; zero ads.
Bodhami AI	Requires trained counsellors; not self-service for students.	Fully self-directed 24/7; instant 30-sec stream matcher.
Jnanabhumi / NSP	Scholarship-only portals; complex navigation; no career tools.	Bilingual unified hub combining careers, books & scholarships.
AICTE Rural Portal	Focuses only on placements; no 10th/Inter roadmaps.	Complete progression ladders from 10th to professional degrees.

Key Gap Identified: No single existing platform integrates curated AI study tools, interactive career roadmaps, 1-click board textbooks, and verified government scholarships into a free, clutter-free bilingual interface for rural students.

VidyaUday vs Competitors

Feature-by-feature comparison — what makes VidyaUday uniquely effective



Key Feature Dimension	Buddy4Study	Careers360	Bodhami	Jnanabhumi	VidyaUday
Curated AI Tools & Copyable Prompts	✗	✗	✗	✗	✓
Interactive Multi-Stream Career Roadmaps	✗	✓	✓	✗	✓
30-Second Scholarship Eligibility Checker	✓	✗	✗	✓	✓
1-Click SCERT Board Textbooks & Papers	✗	✗	✗	✗	✓
Student Scam Buster & Cyber Shield	✗	✗	✗	✗	✓
Authentic Telugu Bilingual Parity	✗	✗	✗	✗	✓
Exclusive Clutter-Free Single-View UX	✗	✗	✗	✗	✓
100% Free Forever (No Login /	✗	✗	✗	✓	✓

Data Gathered / Data Used

Empirical student survey insights combined with verified statutory government data



PRIMARY DATA (Student Survey)

- **Survey Scope:** Field surveys & Google Forms conducted with rural 10th and Inter students in AP & TS.
- **74.2% Guidance Deficit:** Students were unaware of career streams outside MPC/BiPC (such as Polytechnic or Allied Health).
- **81.5% AI Awareness Gap:** Students had never utilized free AI tools for academic doubt-solving or writing help.
- **68.4% Unclaimed Welfare:** Eligible students did not know the criteria or deadlines for state/central scholarships.
- **89.0% Vernacular Demand:** Students requested all guidance and documents in simple native Telugu.



SECONDARY DATA (Verified Sources)

- **Official Government Gazettes:** Curated strictly from AP G.O. Ms. No. 77 & 115 (Jnanabhumi) and TS ePASS CGG regulations.
- **Central Ministry DBT Schemes:** Verified eligibility guidelines from Ministry of Education (NMMS, PM-USP) and AICTE (Pragati/Saksham).
- **Academic Board Curricula:** Official state textbooks, blueprints, and 5-year public papers from AP SCERT and TS SCERT.
- **Strict Data Integrity:** Zero unauthenticated web scraping. All scholarship links redirect strictly to official .gov.in portals.
- **Annual Academic Verification:** Updated specifically for the Academic Year 2024–25 recurring application cycles.



Methodology / System Design — Flow Chart

A structured five-step user journey from confusion to an empowered decision



Step 1:  **Instant Access** — Student opens platform on any phone; selects English or Telugu. No login or installation required.

Step 2:  **Module Selection** — Chooses from AI Tools, Career Guidance, Free Learning Resources, Scholarships, or Digital Skills.

Step 3:  **Single-View Focus** — Exclusive mode switcher opens one view (Roadmaps, Books, or Checker) and hides all other clutter.

Step 4:  **Interactive Action** — Copies AI prompt, compares stream metrics, checks scholarship eligibility, or downloads official PDFs.

Step 5:  **Empowered Outcome** — Directs to authenticated .gov.in portals with verified documents, clear deadlines, and complete confidence.

Methodology / System Design — Block Diagram

Four-tier client-side architecture powering the VidyaUday web application



PRESENTATION LAYER (UI/UX)

Semantic HTML5, CSS3 Custom Properties, Responsive CSS Grid & Flexbox, Dark Glassmorphic Design System, Mobile-First Viewports.

CLIENT LOGIC & REACTIVE STATE LAYER

Vanilla JavaScript (ES6+), Zero Framework Runtime, Isolated Single-View Switchers (switchCareerView, switchResourceView, switchScholView, switchDigitalView).

FUNCTIONAL FEATURE ENGINES

30-Sec Scholarship Eligibility Matcher, Interactive Career Comparison Matrix, Subject Finder, 1-Click Copy Prompt Engine, Digital Readiness Quiz.

VERIFIED KNOWLEDGE & CACHING LAYER

Authenticated Statutory Government DBT Datasets, Bilingual JSON Dictionaries (EN/TE), PWA Service Worker (sw.js) Offline Asset Caching.

Implementation / Core Modules

Core functional modules implemented and client-side reactive eligibility logic

Core Modules Implemented

- **Bilingual Hero Gateway:** Dynamic real-time English/Telugu DOM translation with PWA install support.
- **AI Study Prompts Hub:** Curated engines + 1-click copyable student prompts with live clipboard feedback.
- **Career Guidance Switcher:** 3-view system: Interactive Roadmaps, Stream Comparison, and 30-Sec Matcher.
- **Free Resources Hub:** Top platforms, 1-click SCERT book/paper downloads, and instant subject search.
- **Scholarships Hub:** 20+ verified schemes, 30-sec eligibility checker, and mandatory document guide.
- **Digital Literacy Hub:** Essential online exam skills, 30-sec readiness quiz, and student scam shield.

// Eligibility Matching Algorithm

```
function isScholarshipEligible(s, profile) {  
  const r = s.eligRules;  
  if (!r) return false;  
  // Verify Caste & Social Category  
  if (!r.castes.includes(profile.caste)) return false;  
  // Verify Gender Criteria  
  if (r.gender !== 'all' && profile.gender !==  
    r.gender)  
    return false;  
  // Verify Class & Domicile State  
  if (!r.classes.includes(profile.class)) return false;  
  if (!r.states.includes(profile.state)) return false;  
  // Evaluate Family Income Threshold  
  const sIdx = incomeOrder.indexOf(profile.income);  
  const mIdx = incomeOrder.indexOf(r.incomeMax);  
  return sIdx <= mIdx;  
}
```





Production screens deployed live on GitHub Pages with authentic bilingual UI

VidyaUday
HUB OF KNOWLEDGE

AI Tools Careers Resources Scholarships Coding Digital Skills EN తెలుగు

10th & Intermediate Scholarships Hub

Verified government schemes, instant eligibility check, and essential document guides.

Verified Government Schemes (Academic Year 2024-25): All eligibility criteria are curated from official Government Gazettes and DBT Portals (National Scholarship Portal, AP Jaganabhumi, TS ePASS, AICTE). All application links redirect strictly to official .gov.in websites.

Top Scholarships Hub 30-Sec Eligibility Checker Documents & Portals Guide

All Schemes 10th SSC (Pre-Matric) Intermediate (+2) Andhra Pradesh Telangana Central Govt Girls Only

AP Jagananna Vidya Deevena
STATE GOVERNMENT Annual 2024-25

100% full tuition fee reimbursement credited to mother's bank account quarterly

Eligibility: SC/ST/BC/EBC/Kapu/Minority students in AP; family income ≤ ₹2.5L, studying Inter to PG via Conveyer Quota

Official Source: AP Jaganabhumi (G.O. Ms. No. 77)

Deadline: Oct-Jan annually

AP Jagananna Vasathi Deevena
STATE GOVERNMENT Annual 2024-25

₹10,000-₹20,000/year for hostel and mess charges (ITI: ₹10K, Diploma: ₹15K, Degree/B.Tech: ₹20K)

Eligibility: SC/ST/BC/EBC/Minority students in AP; family income ≤ ₹2.5L for hostel & food expenses

Official Source: AP Social Welfare (G.O. Ms. No. 10)

Deadline: Oct-Jan annually

Telangana ePASS Post-Matric
STATE GOVERNMENT Annual 2024-25

Full tuition fee + monthly maintenance ₹500-₹1,500/month depending on course

Eligibility: SC/ST/BC/EBC/Minority students in Telangana; income ≤ ₹2L, studying Inter to PG

Official Source: TS ePASS (Centre for Good Governance)

Deadline: Sep-Dec annually

Ask Uday!

10th & Inter Scholarships Hub (Verified Govt DBT Schemes)

VidyaUday
HUB OF KNOWLEDGE

AI Tools Careers Resources Scholarships Coding Digital Skills EN తెలుగు

10th & Intermediate Scholarships Hub

Verified government schemes, instant eligibility check, and essential document guides.

Verified Government Schemes (Academic Year 2024-25): All eligibility criteria are curated from official Government Gazettes and DBT Portals (National Scholarship Portal, AP Jaganabhumi, TS ePASS, AICTE). All application links redirect strictly to official .gov.in websites.

Top Scholarships Hub 30-Sec Eligibility Checker Documents & Portals Guide

Your Profiles: Inter 1st Yr Andhra Pradesh BC/OBC ₹1L - ₹2.5L Female Edit Filters

Congratulations! Based on your profile, you qualify for 5 Government Scholarships!

AP Jagananna Vidya Deevena
STATE GOVERNMENT Annual 2024-25

100% full tuition fee reimbursement credited to mother's bank account quarterly

Eligibility: SC/ST/BC/EBC/Kapu/Minority students in AP; family income ≤ ₹2.5L, studying Inter to PG via

AP Jagananna Vasathi Deevena
STATE GOVERNMENT Annual 2024-25

₹10,000-₹20,000/year for hostel and mess charges (ITI: ₹10K, Diploma: ₹15K, Degree/B.Tech: ₹20K)

Eligibility: SC/ST/BC/EBC/Minority students in AP

PM YASASVI Post-Matric (OBC/EBC)
CENTRAL GOVERNMENT Annual 2024-25

Tuition fee reimbursement + academic allowance up to ₹10,000/year

Ask Uday!

30-Sec Eligibility Checker (5 Schemes Matched)

Career Guidance Hub (3-View Switcher & Roadmaps)

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AI Tools Careers Resources Scholarships Coding Digital Skills EN తెలుగు

10th & Intermediate Digital Skills Hub

Master online admissions, DigiLocker certificates, cyber scam safety, and cloud productivity.

Essential Student Skills 30-Sec Digital Readiness Quiz Student Scam Buster Guide

All Skills Govt Portals & Docs Online Exam Forms Cyber Safety Cloud & Productivity

DigiLocker & Legal Digital Records
GOVT OFFICIAL • 100% LEGAL

Legally valid across all AP & TS colleges under IT Act 2000 — never carry fragile paper originals.

- Download official DigiLocker app & link with Aadhaar-registered mobile number
- Push Class 10 (SSC) and Inter marks memos instantly using Board name & Roll number
- Store Caste, Income & Residence certificates permanently without risking rain/physical damage
- Share digital QR-verified credentials directly during AP/TS EAMCET & POLYCET counseling

Online Exam & Admission Form Mastery
EAMCET • POLYCET • DOST

- Avoid paying ₹100-₹200 to cyber café agents for basic form submissions
- Compress passport photos to 30KB-50KB and signature to 10KB-30KB using free browser tools
- Convert phone snapshots of certificates into clear, single-page PDFs under 200KB
- Double-check your name spelling against 10th memo letter-for-letter before clicking Final Submit
- Download and save the PDF Acknowledgement Receipt with Transaction ID immediately

Cyber Scam & OTP Shield
STUDENT CYBER SAFETY

- Protect yourself and parents from WhatsApp scholarship frauds and fake college job offers
- Golden Rule: Central & State Govt scholarships NEVER demand an application fee or OTP
- Spot fake URLs: Verified portals always end in gov.in or nic.in — avoid .xyz, .top, or .club
- UPI Safety: Entering your UPI PIN always DEPLETS money; you never need a PIN to receive funds
- Report student online cyber fraud immediately on the national helpline 1930 or cybercrime.gov.in

Ask Uday!

Digital Literacy Hub (Essential Skills & Scam Shield)

Impact Assessment

Impact Assessment

Social, technological, and community impact across rural student populations



Social Impact

- Bridges the rural guidance gap between private coaching and government schools.
- Delivers complete vernacular equality via authentic Telugu translations.
- Empowers first-generation learners to make confident educational choices.
- Fosters early female education via dedicated girls' scholarship tracks.

Technological Impact

- Lightweight architecture (<450 KB) boots in under 1.2s on 3G mobile networks.
- Zero-framework Vanilla JS eliminates 300KB+ of parser freeze on budget phones.
- Exclusive single-view switcher prevents visual overwhelm and cognitive fatigue.
- PWA service worker caches essential links and documents for offline access.

Community & Economic Impact

- Helps rural families claim ₹10,000–₹50,000+ in annual government DBT scholarships.
- Eliminates unnecessary cyber café commission fees for basic form submissions.
- Broadens career visibility beyond traditional engineering and medical degrees.
- 100% free forever with no registration, no ads, and zero monetization.

Challenges Faced

Key obstacles encountered during development and how each was resolved



1. Statutory Data Verification

⚠ **Challenge:** Navigating scattered, conflicting government circulars with no public REST APIs.

✅ **Solution:** Hand-curated and cross-verified eligibility rules directly against official State Gazettes (AP G.O. Ms. No. 77) and DBT guidelines.

2. Cognitive Clutter Elimination

⚠ **Challenge:** Presenting vast amounts of roadmaps and resources without overwhelming school students.

✅ **Solution:** Engineered an 'Exclusive Single-View Switcher' where selecting one mode hides all others, keeping interfaces bite-sized.

3. Low-Bandwidth Performance

⚠ **Challenge:** Heavy frontend frameworks (React/Vue) stalling on low-cost Android phones on village 3G.

✅ **Solution:** Built entirely with modular Vanilla JS and CSS custom properties, achieving sub-1.2s loads on mobile.

4. Authentic Vernacular Translation

⚠ **Challenge:** Automated machine translation tools produced unnatural and confusing Telugu educational terms.

✅ **Solution:** Created hand-crafted bilingual JSON dictionaries with verified, natural Telugu educational terminology.

Conclusion

What VidyaUday proves and what it delivers for rural students



- **Unified Guidance Ecosystem:** Successfully integrated AI study tools, career roadmaps, board textbooks, and scholarships onto one seamless platform.
- **Linguistic Equity:** Delivered 100% authentic bilingual parity (English + Telugu), ensuring language is never an obstacle to quality education.
- **High-Performance Engineering:** Proved that a clean Vanilla JS architecture outperforms heavy frameworks for rural low-bandwidth users.
- **Zero Financial Barrier:** Operates completely free of cost with zero login barriers, zero ads, and strict respect for user privacy.
- **Measurable Community Impact:** Transforms information fragmentation into structured confidence, enabling students to claim legal government aid.

This project demonstrates how purposeful software engineering can democratize educational opportunity and empower every rural student to achieve their full potential.

Future Work

Planned enhancements to scale VidyaUday across Andhra Pradesh & Telangana



Automated WhatsApp & SMS Reminders

Integrate opt-in SMS/WhatsApp alerts notifying rural parents and students 7 days before major scholarship portal closures (AP ePASS, NSP, NMMS).

Vernacular Voice AI Assistant

Implement Web Speech interactive voice queries in Telugu, allowing students with lower reading fluency to ask questions orally and listen to spoken answers.

Downloadable Offline Study Packs

Build one-click zipped revision packages containing formula cheat sheets, blueprints, and 5-year solved board papers for complete offline village study.

Rural Peer Mentorship Network

Create an anonymous, verified mentorship channel connecting high school students with rural alumni currently excelling in government engineering & medical colleges.

References

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